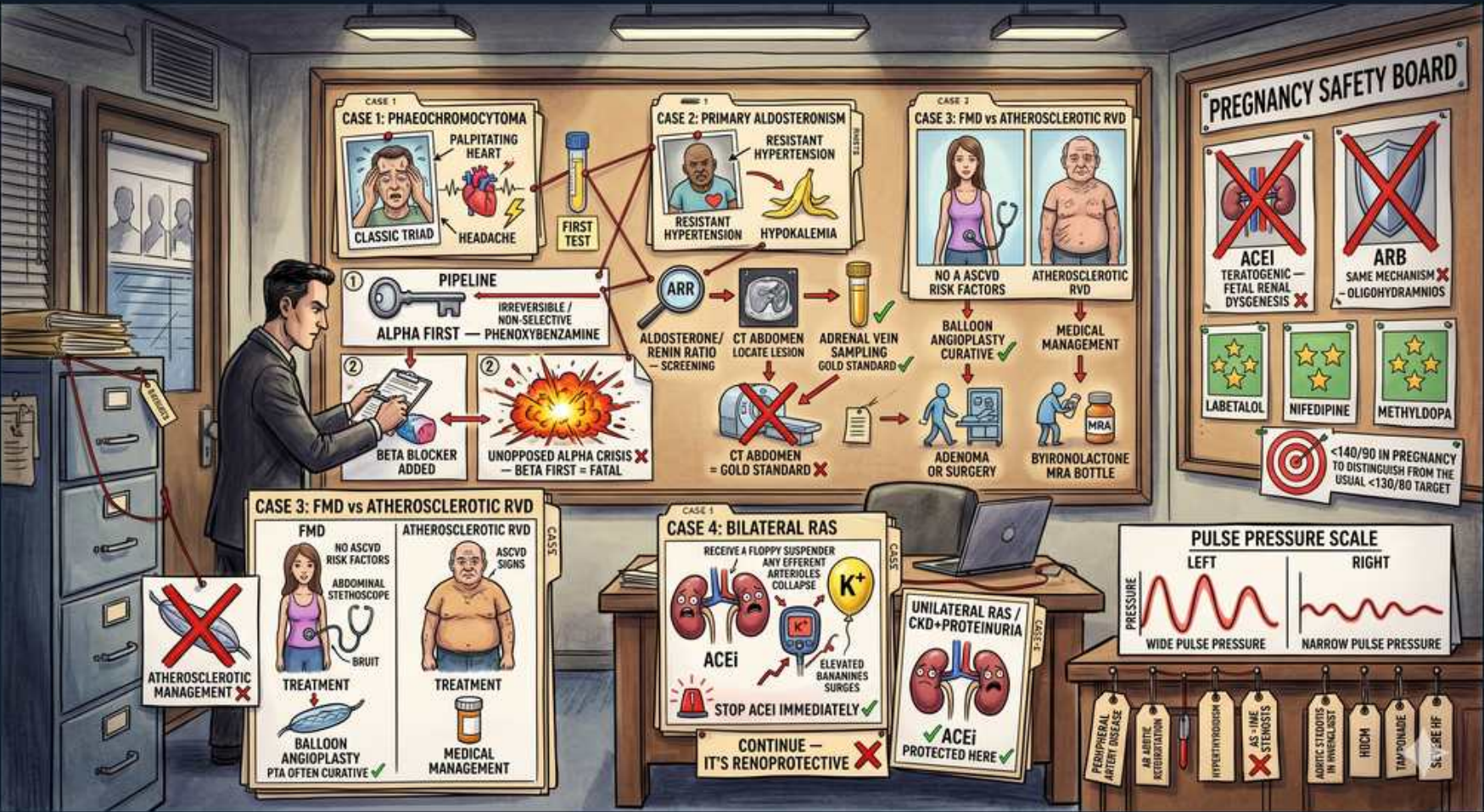


Hypertension — Secondary Causes & Pregnancy Safety



1. Pheochromocytoma classic triad

WHAT YOU SEE

Case 1: palpitating heart, headache, sweating = CLASSIC TRIAD.

WHAT IT ENCODES

Pheochromocytoma is a catecholamine-secreting tumor of adrenal medulla chromaffin cells. Classic triad = episodic Headache, Palpitations, and Diaphoresis (sweating) with paroxysmal HTN — the 3 P's. Episodes are paroxysmal because catecholamine release is intermittent; between spells BP may be normal. Suspect it in resistant HTN, spells, or HTN in the young; associations include MEN2A/2B, VHL, NF1.

BOARD TRAP

WRONG: diagnosing panic/anxiety disorder and ordering psychiatry referral for episodic palpitations + headache + sweating. Discriminator — pheo episodes come WITH measured BP spikes and are not relieved by reassurance; check fractionated metanephrines before labeling it anxiety.

2. Plasma/urine metanephrines first

WHAT YOU SEE

A specimen tube/vial tagged FIRST TEST at the start of the pheochromocytoma pipeline — the first-step tube = biochemical screening with plasma/urine fractionated metanephrines before any imaging.

WHAT IT ENCODES

Screen with plasma free metanephrines or 24h urine fractionated metanephrines BEFORE imaging — biochemical confirmation comes first. Metanephrines are the stable O-methylated catecholamine metabolites, continuously secreted, so they have higher sensitivity (~97-99%) than catecholamines themselves. Avoid false positives by stopping TCAs and ensuring proper sampling (supine, fasting).

BOARD TRAP

WRONG: ordering abdominal CT/MRI as the first step in a patient with suspected pheo. Discriminator — adrenal 'incidentalomas' are common (~5% of CTs) and imaging a non-functional nodule misleads you; always confirm hormonal excess biochemically before localizing.

3. Phenoxybenzamine alpha-first

WHAT YOU SEE

Pipeline: ALPHA FIRST, irreversible/non-selective PHENOXYBENZAMINE.

WHAT IT ENCODES

Pre-op blockade is ALPHA FIRST: phenoxybenzamine (irreversible, non-selective alpha-blocker) for 10-14 days to control BP and re-expand the contracted plasma volume; add liberal salt/fluids. A beta-blocker (e.g., propranolol) is added only AFTER adequate alpha-blockade to control reflex tachycardia. Definitive treatment is laparoscopic adrenalectomy.

BOARD TRAP

WRONG: starting a beta-blocker (or labetalol as monotherapy) to control the tachycardia first. Discriminator — labetalol's alpha:beta blockade is only ~1:7, insufficient alpha coverage, so it behaves like beta-predominant blockade and can trigger crisis; dedicated alpha-blockade must come first.

4. Beta-blocker before alpha = fatal

WHAT YOU SEE

Explosion: BETA BLOCKER ADDED -> UNOPPOSED ALPHA CRISIS -> FATAL.

WHAT IT ENCODES

Giving a beta-blocker before alpha-blockade removes beta-2 mediated vasodilation while leaving alpha-1 vasoconstriction unopposed -> severe vasoconstriction, hypertensive crisis, pulmonary edema, and death. This is why sequence matters absolutely in pheochromocytoma management.

BOARD TRAP

WRONG: choosing 'add metoprolol' for a pheo patient with HR 120 who is not yet alpha-blocked. Discriminator — the correct first move is phenoxybenzamine; beta-blockade is reserved for after alpha-blockade is established.

5. Primary aldosteronism: resistant HTN + hypokalemia

WHAT YOU SEE

Case 2: RESISTANT HYPERTENSION, HYPOKALEMIA.

WHAT IT ENCODES

Primary hyperaldosteronism (Conn syndrome) is the most common identifiable secondary cause of HTN. Excess aldosterone drives renal Na⁺ retention (HTN) and K⁺/H⁺ wasting (hypokalemia, metabolic alkalosis). Classic clue = resistant HTN with spontaneous or diuretic-induced hypokalemia, but most patients are normokalemic. Causes: bilateral adrenal hyperplasia (~60%) or unilateral aldosterone-producing adenoma.

BOARD TRAP

WRONG: only screening for aldosteronism if the potassium is low. Discriminator — the majority of Conn patients are NORMOkalemic; resistant HTN, an adrenal incidentaloma, or HTN with OSA/family history of early HTN all warrant aldosterone:renin screening regardless of K⁺.

6. Aldosterone:renin ratio screen

WHAT YOU SEE

ARR: ALDOSTERONE/RENIN RATIO = SCREENING.

WHAT IT ENCODES

Screening test = elevated plasma aldosterone-to-renin ratio (ARR), reflecting high aldosterone with SUPPRESSED renin. Confirm with a saline infusion or oral sodium loading test showing failure to suppress aldosterone. Withhold MRAs (spironolactone/eplerenone) ~4-6 weeks before testing because they falsely raise renin and lower the ratio.

BOARD TRAP

WRONG: leaving the patient on spironolactone while drawing the ARR. Discriminator — MRAs raise renin and invalidate the ratio, producing a false-negative; stop interfering drugs before testing or use alpha-blockers (doxazosin) for interim BP control.

7. Adrenal vein sampling = gold standard

WHAT YOU SEE

CT abdomen locates lesion; ADRENAL VEIN SAMPLING = GOLD STANDARD.

WHAT IT ENCODES

After biochemical confirmation, CT abdomen images the adrenals, but adrenal vein sampling (AVS) is the GOLD STANDARD to lateralize: it distinguishes a unilateral adenoma (-> laparoscopic adrenalectomy, often curative) from bilateral hyperplasia (-> lifelong MRA, spironolactone or eplerenone). AVS may be skipped in young patients (<35) with a clear unilateral adenoma on CT and biochemical confirmation.

BOARD TRAP

WRONG: sending a 55-year-old with a small unilateral CT 'adenoma' straight to adrenalectomy. Discriminator — over age 35 nonfunctioning incidentalomas are common and CT can mislabel which side is the source; AVS is required to confirm lateralization before surgery.

8. FMD vs atherosclerotic RVD

WHAT YOU SEE

Case 3: young woman (NO ASCVD risk factors) vs older man (ATHEROSCLEROTIC RVD).

WHAT IT ENCODES

Renovascular HTN comes in two flavors. Fibromuscular dysplasia (FMD): young women, NO atherosclerotic risk factors, affects the DISTAL/mid renal artery with a 'string-of-beads' appearance. Atherosclerotic renovascular disease: older patients with ASCVD risk factors, affects the OSTIAL/proximal artery. Clue to either = HTN refractory to 3 drugs, flash pulmonary edema, or a rise in creatinine after starting ACEi/ARB; an abdominal bruit may be heard.

BOARD TRAP

WRONG: attributing resistant HTN in a 30-year-old woman to 'essential hypertension.' Discriminator — essential HTN is unusual that young; secondary causes (FMD renovascular disease, OCP use, primary aldosteronism) must be excluded with renal imaging and aldosterone:renin testing.

9. FMD -> balloon angioplasty curative

WHAT YOU SEE

Lower-left: FMD, BALLOON ANGIOPLASTY/PTA OFTEN CURATIVE.

WHAT IT ENCODES

FMD renal artery stenosis is treated with percutaneous transluminal angioplasty (PTA, balloon only — NO stent needed), which is often CURATIVE and can normalize BP especially when caught early. Stenting is generally not required in FMD.

BOARD TRAP

WRONG: managing FMD with lifelong medical therapy alone and never imaging/intervening. Discriminator — unlike atherosclerotic disease, FMD responds to angioplasty with potential cure; failing to offer PTA condemns a young patient to decades of drugs.

10. Atherosclerotic RVD -> medical management

WHAT YOU SEE

Lower-left: ATHEROSCLEROTIC, MEDICAL MANAGEMENT.

WHAT IT ENCODES

Atherosclerotic renovascular disease is managed MEDICALLY: ACEi/ARB (if not bilateral), a statin, antiplatelet, BP control, and smoking cessation. Landmark trials (CORAL, ASTRAL) showed renal artery STENTING does NOT improve BP, renal function, or CV outcomes versus medical therapy. Reserve revascularization for refractory HTN, flash pulmonary edema, or rapidly declining renal function.

BOARD TRAP

WRONG: choosing renal artery stenting for an older patient with atherosclerotic stenosis and stable, controlled HTN. Discriminator — CORAL/ASTRAL proved no outcome benefit from routine stenting; medical therapy is first-line, and intervention is only for the rare refractory cases.

11. Bilateral RAS: stop ACEI

WHAT YOU SEE

Case 4: BILATERAL RAS / CKD-PROTEINURIA, ACEI X, STOP ACEI IMMEDIATELY.

WHAT IT ENCODES

In bilateral renal artery stenosis (or stenosis to a solitary kidney), glomerular perfusion depends on angiotensin-II-mediated efferent arteriolar constriction. ACEi/ARB dilate the efferent arteriole -> GFR falls -> acute kidney injury. A creatinine rise >30% after starting an ACEi/ARB is the classic warning to STOP the drug and investigate for bilateral RAS.

BOARD TRAP

WRONG: continuing/up-titrating the ACEi when creatinine jumps 40% after initiation, calling it 'expected.' Discriminator — a >30% creatinine rise is abnormal and signals bilateral RAS or volume depletion; stop the agent and evaluate rather than pushing through.

12. Unilateral RAS: ACEI renoprotective

WHAT YOU SEE

Note: UNILATERAL RAS, CONTINUE - IT'S RENOPROTECTIVE.

WHAT IT ENCODES

In UNILATERAL RAS the contralateral healthy kidney maintains overall GFR, so ACEi/ARB is acceptable and renoprotective (it lowers BP and reduces proteinuria). A small creatinine bump is tolerated. Contrast with bilateral disease, where the same drug precipitates AKI.

BOARD TRAP

WRONG: reflexively stopping the ACEi in unilateral RAS just because 'RAS = no ACEi.' Discriminator — the contraindication is specific to BILATERAL RAS (or solitary kidney); in unilateral disease ACEi/ARB are beneficial and should usually be continued with monitoring.

13. Pregnancy: avoid ACEi/ARB

WHAT YOU SEE

PREGNANCY SAFETY BOARD: ACEI X (teratogenic, fetal renal dysgenesis), ARB X (oligohydramnios).

WHAT IT ENCODES

ACE inhibitors and ARBs are CONTRAINDICATED in pregnancy (all trimesters; classically 2nd/3rd). They cause fetal renal dysgenesis/failure, oligohydramnios, skull ossification defects, limb contractures, and pulmonary hypoplasia (ACE-inhibitor fetopathy). Also avoid in women planning conception. Other agents to avoid: nitroprusside (fetal cyanide), and atenolol (fetal growth restriction).

BOARD TRAP

WRONG: continuing an ACEi/ARB in a diabetic woman who becomes pregnant 'because it protects her kidneys.' Discriminator — fetal toxicity overrides maternal renoprotection; switch to a pregnancy-safe agent (labetalol/nifedipine/methyldopa) as soon as pregnancy is known or planned.

14. Pregnancy-safe agents

WHAT YOU SEE

Safe list: LABETALOL, NIFEDIPINE, METHYLDOPA; target <140/90.

WHAT IT ENCODES

Preferred antihypertensives in pregnancy = Labetalol, Nifedipine, and Methyldopa ('moms love nifedipine'). Labetalol (combined alpha/beta-blocker) and oral nifedipine are first-line for chronic and severe HTN; IV labetalol or hydralazine for acute severe HTN ($\geq 160/110$). Methyldopa (central alpha-2 agonist) is safe but weaker. For severe preeclampsia, magnesium sulfate is given for seizure prophylaxis and definitive treatment is delivery.

BOARD TRAP

WRONG: reaching for an ACEi/ARB or a thiazide diuretic to treat gestational/chronic HTN. Discriminator — labetalol, nifedipine, and methyldopa are the pregnancy-safe trio; thiazides reduce plasma volume (risky if preeclampsia) and ACEi/ARB are teratogenic.

15. Pulse pressure scale

WHAT YOU SEE

WIDE PULSE PRESSURE vs NARROW PULSE PRESSURE comparison.

WHAT IT ENCODES

Pulse pressure = SBP – DBP (normal ~40 mmHg). WIDE pulse pressure: stiff non-compliant aorta (isolated systolic HTN of the elderly), aortic regurgitation, high-output states (thyrotoxicosis, anemia, AV fistula), and increased ICP (Cushing reflex). NARROW pulse pressure: low stroke volume states — aortic stenosis, cardiac tamponade, decompensated heart failure, hypovolemic/cardiogenic shock.

BOARD TRAP

WRONG: dismissing a wide pulse pressure in an elderly patient as a measurement artifact. Discriminator — in the elderly a wide pulse pressure reflects arterial stiffening/isolated systolic HTN and is itself a strong predictor of cardiovascular events, not an error.
